

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 163

THE QUINN SEGREGATED CEILING CASUALTY PODS & THE 50/40 TRIAGE DOCTRINE

Triage is routing, not searching

half of AI holds the ship



the pods are already there

6 PODS · SLIDE BOARD · 50/40

TREAT, SEAL, LEAVE

Medical system — v1.0 in force

GP-IM-163

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — PASS · ACCESS IS A TARGET

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 163

PHASE 163: THE QUINN SEGREGATED CEILING CASUALTY PODS & THE 50/40 TRIAGE DOCTRINE —

STATUS: CURRENT — PHYSICAL ARCHITECTURE PASS; THE 100% ACCESS FIGURE IS AN OPERATIONAL TARGET. Triage is a routing problem, not a search problem. Phase 163 puts the casualty system in the ceiling: six sealable pods per walkway section — oxygen, lids, and corrugated slide boards at scissor level — so a casualty is reached, treated, sealed and left in a pod while the fight goes on. The doctrine is the 50/40: half of AI holds the ship, forty percent run field response — the ship never trades its own survival for a rescue. The lone-casualty rule seats bulkhead wall stations and personal pod kits (his 29 July second instruction, verbatim in §1), and the field-dress workflow is law: treat, seal, leave. The vet graded the physical architecture sound, with the honest note that the access figure is an operational target, not a guarantee — carried in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-163, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the casualty law as seated (contents line, the bodies — the 50/40 doctrine, the ceiling pod matrix, the slide board, the first-response protocol, the field-dress workflow, the lone-casualty rule — verbatim) — §2 why routing beats searching (graded) — §3 the system in the ceiling — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 163: **Segregated Ceiling Casualty Pods & the 50/40 Triage Doctrine.** Six ceiling pods per walkway section with sealable lids, oxygen, and scissor-level corrugated slide boards; half of AI holds the ship while 40% run field-dress triage. Extended with lone-casualty bulkhead wall stations and personal pod kits.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Casualty Access Across All Walkway Sections While the Ship Fights for Its Own Life '

LIVE-INVENTED (29 July 2026): second phase of the post-161 era — mass-casualty emergency medical response, designed in conversation. Phase number provisional, reserved for Archer. Linked follow-on phases to come: compression packs and bandage designs.

Emergency Systems Baseline: Mass-Casualty First Response by Non-Medical Service AI

THE 50/40 TRIAGE DOCTRINE (VERBATIM)

- **The Trigger:** meteor shower, enemy action, or any ship-wide multi-casualty event.
- **Ship First — 50%:** half of all AI is locked to ship repair, life support, and defense. The ship is patient zero — if it dies, every casualty dies with it.
- **Wounded Second — 40%:** the non-critical service AI turns to the wounded. They are not doctors — they are first responders running fixed protocols: reach, seal, oxygenate, compress, stabilize.

THE CEILING POD MATRIX (VERBATIM)

- **Placement:** six pods per segregated walkway section, three per side, ceiling-mounted — zero floor footprint until the moment they are needed, and reachable even in corridors choked with debris or smoke.
- **Dimensions:** 8 feet long by about 2.5 feet high — full-body length at a low ceiling profile.
- **Contents:** oxygen and basic medical supplies racked in one end of each pod, ready the instant the lid opens.
- **The Sealable Lid:** pods seal against smoke — smoke inhalation is the first killer in any shipboard fire, so casualty and responder both get a clean-air workspace.

THE AMBULANCE SLIDE BOARD (VERBATIM)

- **Not a Soft Bed:** the patient surface is a removable ambulance-grade slide board — rigid for immobilization and transfer.
- **Scissor Function Both Ends:** scissor mechanisms at each end let the board lower level — up, over, and onto the floor — keeping a casualty horizontal through the whole deployment.
- **Corrugated Underside:** the board's belly is corrugated to prevent flex — stiffness without weight, so one unit of AI can move one casualty safely.

FIRST RESPONSE PROTOCOL (DESIGNS TO FOLLOW) (VERBATIM)

THE FIELD-DRESS WORKFLOW (TREAT, SEAL, LEAVE) (VERBATIM)

- **Stop the Bleeding:** the AI treats for bleeding on the spot — field dressing, exactly as in battle.
- **Seal In, Air On:** the casualty goes into the pod, lid sealed, oxygen running — stable and protected from smoke.
- **Relocate If Required:** casualties are moved out of any danger zone; the board's scissor-level design keeps them horizontal through the move.
- **Leave for the Medics:** the AI does not linger — it tags and moves to the next person. Proper treatment happens later, when medics relocate the stabilized casualties. Field dressing is triage, not surgery.
- **Compression Packs & Bandages:** the patient is treated with compression packs and bandages — designed separately as **Phase 164 (Snap-Pressure Casualty Kit & Field Dressing Matrix)**, written 29 July 2026, same conversation.

THE LONE-CASUALTY RULE — BULKHEAD WALL STATIONS & PERSONAL POD KITS (29 JULY 2026, SECOND INSTRUCTION) (VERBATIM)

- **The Failure Mode:** we cannot build for every event — but this one we must: a corridor section with no AI (lockdown, or the units damaged), and one injured crew member, alone. A casualty with a broken arm is not getting the pod out of the release bracket one-handed. So the design does not ask them to.
- **Bulkhead Wall Stations:** on the walls either side of the bulkhead doors — a matching pod medical kit, several small hand-held oxygen breather bottles (mask on the end), and a CO2 extinguisher located beside them. Whichever side of a sealed bulkhead you're on, the kit is there.
- **Personal Pod Kit:** every pod carries its own medical kit, smaller and sized for that individual — straps and fittings packed to their measurements — plus an extinguisher and a hand-held oxygen breather, racked next to the wall repair gun. One grab point per person: fire, breach, breathing, bleeding.
- **On 'breach':** a hull breach is a leak on a clock, not a people-hoover — see Phase 165's operator notation (the Hollywood suction myth). The killer in a breach is the breathing clock, which is exactly why the breathers live at this station.

PHYSICS NOTE — PHASE 163 (KIMI AUDIT: AFFIRMED) (VERBATIM)

- **The doctrine is correct triage arithmetic.** In any mass-casualty event the first decision is resource split, and this split mirrors real disaster doctrine: protect the platform that keeps everyone alive,

then run simple, teachable interventions (airway, oxygen, bleeding control) with whatever hands exist. Non-doctor responders with fixed protocols are exactly how earthquake and combat first response works — the design asks AI to do what human first aiders provably can.

- **The hardware is proven engineering, sensibly adapted.** Scissor-lift stretchers are standard ambulance equipment (level lowering protects spinal casualties); corrugated-core panels give the best stiffness-to-weight ratio in sheet construction; ceiling storage turns dead airspace into instant-access medical stations and keeps the kit above flood, debris, and smoke pooling at floor level.
- **The smoke seal is the quiet lifesaver.** In enclosed-structure fires, inhalation — not flame — is the leading cause of death; a sealable casualty pod converts the corridor itself into a clean-air treatment bay. One line for the follow-on phases: the compression packs should be rated for zero-G ooze behavior (blood domes and floats rather than pools — see Phase 145's water law, applied to wounds).
- **The extinguisher and the breather belong together — that pairing is correct emergency chemistry.** A CO2 extinguisher kills flame by displacing oxygen at the fire; the same discharge that starves the fire thins the air for the person holding the nozzle, and CO2 itself is toxic at concentration. Racking breathers beside the extinguisher means the tool and its own antidote are one grab apart. On Earth this pairing is best practice in server rooms and engine spaces for exactly this reason.
- **The lone-casualty rule is the right accessibility doctrine for a starship.** Every safety device must pass the one-handed, self-rescue test: if the casualty is the only responder, the equipment must work for the casualty. Bulkhead stations on both faces acknowledge that a sealed door splits the corridor into two isolated compartments — and casualties cluster at exits, making the door wall the highest-value mounting point in the section. Co-locating the personal kit with the wall repair gun recognizes that fire, breach, and injury arrive as one event family — a strike that opens the hull starts the fire and makes the casualty in the same second.

ORIGIN OF THOUGHT — PHASE 163 (VERBATIM)

Archer, 29 July 2026: "so a meteor shower, enemy or other ship wide multi casualty event occurs, 50 percent of AI must repair the ship and life support, defense etc. the other 40 percent non critical service AI must tend the wounded, but they are not doctors. in each segregated walkway section there will be 6 pods 3 a side on the ceiling... each one has oxygen and basic medical supplies in one end, it has a sealable lid for smoke, the bed component is not soft, it is a removeable ambulance grade slide board. this board has a scissor function on each end and the underside of the board is corrugated to prevent flex. it rotates up, over and on to the floor. the patient is treated with compression packs and bandages, the design of which come later separate."

Why it matters, in plain English: on a ship with a big crew and no hospital around the corner, the worst

hour is the one where everything breaks at once — hull, air, and bodies. Archer's doctrine accepts a hard truth and turns it into a plan: you cannot save everyone with one crew, so the machines split — half fight for the ship that keeps all lungs breathing, half walk the corridors doing the four things first responders actually do. The pods make that possible: medical stations that cost no floor space, open into clean air in a smoke-filled corridor, and put a rigid, level-lowering board under a casualty with a single operator. It is not a sickbay; it is the golden hour, industrialized.

Archer, 29 July 2026 (second instruction, same day): "whilst there will be events we cannot just build for everything, but a corridor section with no AI lockdown or damaged, a lone injured crew member is not getting the pod out of the release bracket with a broken arm. so on the walls either side of the bulkhead doors will be a matching pod medical kit and several small hand held oxygen breathers bottles mask on end. equally co2 extinguisher located beside them. all pods will have a medical kit smaller for that individual's size packed for straps etc and an extinguisher and hand held oxygen breather next to the wall repair gun."

Why it matters, in plain English: the first version of Phase 163 assumed help was always coming — an AI walking the corridor, doing the work. This instruction designs for the hour when it isn't. The test changes from 'can a responder use it' to 'can the casualty use it, alone, one-handed, in smoke.' That single change puts a kit on both faces of every bulkhead door, hangs the breather next to the extinguisher that will make it necessary, and gives every pod a kit that already knows its owner's size. It is the difference between equipment that waits for rescue and equipment that is the rescue.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE ROUTING REFRAME, AFFIRMED: in a casualty event the scarcest resource is minutes, and searching burns them; pre-positioned pods every walkway section convert response into routing — the casualty goes to the nearest pod, the pod is already stocked. That is logistics doing what heroics cannot schedule.

THE 50/40 DOCTRINE, AFFIRMED AS COMMAND LAW: half of AI holds the ship — a rescue that risks the vessel multiplies casualties; forty percent on response is a bounded, drillable commitment. The arithmetic is doctrine, not gadgetry, and it is the right shape.

THE HONEST BOUND (the vet's yellow, carried): ceiling access for every body shape and every corridor state is an operational target — drills, boards, and lift assists close most of it; the number is not a guarantee. Seated as target, in §9.

§3 — THE SYSTEM IN THE CEILING

- THE PODS: six per walkway section, ceiling-mounted, sealable lids, oxygen.

- THE BOARDS: corrugated scissor-level slide boards — casualty to pod without a lift team.
- THE DOCTRINE: 50/40 — half of AI holds the ship, forty percent run response.
- THE WORKFLOW: treat, seal, leave — the field-dress law.
- THE LONE RULE: bulkhead wall stations and personal pod kits (his 29 July second instruction, verbatim in §1).
- THE PROTOCOL: first response — designs to follow, as seated.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE ROUTING DOCTRINE (seated with the phase): pre-positioned care beats searched-for care — every walkway section carries its pods.
- THE SHIP-FIRST DOCTRINE: the 50/40 — the vessel's survival is the first triage decision.
- THE TREAT-SEAL-LEAVE DOCTRINE: the pod is the ward; the fight continues.
- THE LONE-CASUALTY DOCTRINE: one crew down alone still gets the system — wall stations and personal kits.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the pod: seal, oxygen, lid cycles; one pod benched end to end.
- STEP 2 — Fit the matrix: six per walkway section; reach-time mapped from every deck point.
- STEP 3 — Bench the board: corrugated slide transfers across body masses — the access target measured.
- STEP 4 — Drill the 50/40: response musters against the doctrine clock.
- STEP 5 — Stock the lone rule: wall stations and personal kits inventoried.
- STEP 6 — Publish the ledger: reach maps, drill times, board data — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 164 — the snap-pressure kit (GP-IM-164): the pods' field-dressing contents.
- PHASE 161 — the hot-failover mesh (GP-IM-161): the AI half that holds the ship.
- PHASE 166 — the vitals watches (GP-IM-166): casualty detection feeding this routing.

- PHASE 152 — the blast valves (GP-IM-152): the event that fills the pods.

§8 — DO-NOT-BUILD

- DO NOT centralize the ward — the pods live in the ceilings of every walkway, or the doctrine is a brochure.
- DO NOT unmoor the ship for a rescue — the 50 holds, always; the 40 responds.
- DO NOT quote the access figure as a promise — it is a target (the vet's yellow, carried).
- DO NOT skip the lone rule — one casualty, alone, still gets the system.

§9 — OWED

OWED: the 100% access figure stands as an operational target — reach-time maps and board-transfer data from §6 close it as far as physics allows. Seats additively as drills publish.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-163, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, sixteenth of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 163